Problem A. Great Golden 'Graph

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Amidst the dust of the Wild West, the whispered rumors of gold spread to your sparse frontier town. And with it, trouble. All your life, you've been tending your herd of cattle, but lately, wanted posters have been posted everywhere. The latest danger has been an outlaw named the Three-Legged Trigger, a bandit who has escaped the law for years now.

In the distance, the old deputy sheriff rides towards you and hands you a mysteriously-encoded telegraph. Is it a new report on the Three-Legged Trigger, or something else?

The telegraph is a string s and a positive integer k. If the string s is repeated over and over again, what is the kth character that appears?

Input

The first line of the input contains a string s ($1 \le |s| \le 10^4$) — the telegraph string. s contains only lowercase English letters and digits 0-9.

The second line of the input contains a single integer k ($1 \le k \le 10^9$) — the position of the character we want.

Output

Output the character that appears at position k if s is repeated over and over again.

Examples

standard input	standard output
`pclassicspring2024` `24`	`s`
`wildwildwest` `160`	`d`

Note

We see that if we repeat "`pclassicspring2024`" over and over again, the 24th character is `s`.

`pclassicspring2024pclass<u>s</u>icspring2024pclassicspring2024...`

The 160th character when "`wildwildwest`" is repeated is a `d`.

Problem B. Cowculated Cattle Combining

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Once you are back at your ranch, you decode the telegraph. The telegraph confirms the rumors — there is gold and lots of it! Your distant cousin tells you about a secret gold mine far away in the mountain ranges of California. You plan to start your long journey once you lasso in your three remaining cattle.

There are three cattle, each marked with an integer a, b, c. You have a multiplication lasso and a subtraction lasso, and you can use each lasso exactly once. The multiplication lasso allows you to choose any two cattle to lasso and a new cattle will replace the two cattle with the product of the two cattle. For example, lassoing cattle a and b will result in a single cattle $(a \cdot b)$. The subtraction lasso allows you to choose any two cattle to lasso and a new cattle will replace the two cattle with the difference of the two cattle. For example, lassoing cattle a and b will result in a single cattle $(a - b)$.

Your challenge is to determine whether it's possible, with 1 multiplication lasso and 1 subtraction lasso, to end up with a cattle marked with the value of 0. Note that you can use either the multiplication lasso or the subtraction lasso first and that a lasso can be applied to any two cattle (not necessarily adjacent).

Input

The first line contains three integers a, b, and c ($-10^4 \le a, b, c \le 10^4$) — the markings on the cattle.

Output

Output "`YES`" if it is possible to use the lassos to get a cattle marked with the value of 0, and "`NO`" otherwise.

You can output the answer in any case (upper or lower). For example, the strings "`yEs`", "`yes`", "`Yes`", and "`YES`" will be recognized as positive responses.

Examples

standard input	standard output
`5 10 2`	`YES`
`1 3 2`	`NO`
`2 10 0`	`YES`

Note

It is possible in the first case. First we can use the multiplication lasso on 5 and 2 to get a 10. Then we can use the subtraction lasso on 10 and 10 to get a 0.

It can be shown that in the second case, it is impossible to get a 0.

Problem C. Shooter Sheriff Showdown

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Before heading out, the veteran sheriff gives you some information on the Three-Legged Trigger. He's called the Three-Legged Trigger because he stamps his crime scenes with three footprints. His revolver is said to be made out of gold, and he has the fastest trigger in the West.

While the Three-Legged Trigger has never been caught, he has a gang of bandits who have had encounters with the sheriff's deputies in the past.

There have been n encounters between the sheriff's deputies and the Three-Legged Trigger's gang in the past. Out of the n encounters, the sheriff's deputies have won x of them. Each encounter results in either the deputies winning or the gang winning (there are no ties). What is the minimum win streak that Three-Legged Trigger's Gang could have had?

Input

The first line contains two integers n and x ($1 \le n \le 10^9$, $0 \le x \le n$) — the number of games played and the number of games that the deputies have won.

Output

Output a single integer, the minimum win streak that Three-Legged Trigger's Gang could have had.

Examples

standard input	standard output
4 1	2
10 2	3
5 5	0
1 0	1

Note

In the first case, the minimum win streak that the gang could have had is 2. This could be accomplished by `GDGG`, where `G` is gang wins and `D` is deputy wins.

In the second case, the minimum win streak that the gang could have had is 3. This could be accomplished by `GGGDGGDGGG`.

In the third case, the gang won no encounters, so the minimum win streak is 0. This could have been accomplished by `DDDDD`.

Problem D. Fast and Ferryous

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

You wave goodbye to the sheriff and make your way to the ferry station. The fastest path westward is by ferry, and you arrive at the port. Unfortunately, there's a long line of people waiting to get on the ferry as well, and the ferry operator is missing! It looks like the only way west is if you operate the ferry yourself.

There are n people who want to board the ferry, and t towns along the river that the ferry travels. Each person has a destination town in mind. The ferry has a maximum carrying capacity of c people. The ferry takes t_f time to move from one town to another and 0 time to let people off. By not taking the ferry, each person takes t_w time to move to the next town on foot. The ferry **can only travel in one direction**, westward. More formally, the ferry visits towns in increasing order, starting from town 0. You want to minimize the sum of times it takes each person to get to their desired town. Find that minimum time.

Input

The first line contains n ($1 \le n \le 1000$) — the number of people.

The second line contains t ($1 \le t \le 1000$) — the number of towns.

The third line contains n integers $d_1, d_2, \ldots, d_n$ ($1 \le d_i \le t$) — the destination towns of the n people.

The fourth line contains c ($1 \le c \le n$) — the capacity of the ferry.

The fifth line contains t_f ($1 \le t_f \le 100$) — the time it takes for the ferry to move from one town to another.

The sixth line contains t_w ($t_f \le t_w \le 100$) — the time it takes to move from one town to another on foot.

Output

Output a single integer, the minimum sum of times it takes for each person to get to their desired town.

Examples

standard input	standard output
4 10 4 5 6 7 3 2 10	76
3 3 1 2 3 3 1 10	6
3 3 3 3 3 2 1 10	36

Note

In the first case, the people going to towns 5, 6, and 7 can take the ferry and the person going to town 4 can move by foot.

The person going to town 4 takes $4 \cdot 10 = 40$ time.

The person going to town 5 takes $5 \cdot 2 = 10$ time.

The person going to town 6 takes $6 \cdot 2 = 12$ time.

The person going to town 7 takes $7 \cdot 2 = 14$ time.

All together, the sum of times for this arrangement is $40 + 10 + 12 + 14 = 76$. It can be shown that this is the minimum possible.

Problem E. Sunset Saloon

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

After an exhausting ferry ride, you stop by a dimly lit saloon at sunset. To your surprise, the old deputy sheriff is here! He's been tracking the Three-Legged Trigger, and he's heard news that the outlaw has been around the area. He invites you to a friendly game of cards.

The card game is as follows. There are n cards in your hand, each with a positive integer a_i on the card. You do this once for each card i: take a card with the index of your choosing c_i $(1 \le c_i \le n)$ with value a_{c_i}, and add $|a_i - a_{c_i}|$ to your sum. Find the maximum sum you can be achieve with the cards in your hand. Note that c can have duplicates, so for $i \ne j$ you can have $c_i = c_j$.

Input

The first line contains a single integer n $(2 \le n \le 10^5)$ — the number of cards in your hand.

The second line contains n integers $a_1, a_2, \ldots, a_n$ $(1 \le a_i \le 10^4)$ — the values of the cards.

Output

Output a single integer, the maximum sum that can be achieved with the cards in your hand.

Examples

standard input	standard output
3 1 8 3	19
5 10 11 2 1 15	60

Note

For the first case, we can maximize the sum as follows.

For the card at index $i = 1$, we can choose $c_1 = 2$, and add $|a_1 - a_2| = |1 - 8| = 7$ to our sum.

For the card at index $i = 2$, we can choose $c_2 = 1$, and add $|a_2 - a_1| = |8 - 1| = 7$ to our sum.

For the card at index $i = 3$, we can choose $c_3 = 2$, and add $|a_3 - a_2| = |3 - 8| = 5$ to our sum.

Overall, our sum is $7 + 7 + 5 = 19$. This can be shown to be the maximum sum possible with this hand.

Problem F. Stupid Slow Stagecoaches

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

The next morning, you thank the sheriff for his company last night. He laughs and says his gold star seemed to do most of the talking. You decide to hitch a ride on a stagecoach further west. Still, the expressions of all the travelers are tense. As of late, the Three-Legged Trigger's Gang has been robbing stagecoaches, and the narrow path ahead through the canyons is ripe for robbery.

There are n stagecoaches all moving west in a line. Stagecoach i starts at a position x_i with speed v_i. If a stagecoach reaches a slower stagecoach in front of it, it will slow down to match the speed of the slower stagecoach. None of the stagecoaches will pass another. As time passes, the stagecoaches will catch up to slower stagecoaches and move together in groups at the same speed. Find the number of stagecoach clumps that eventually form after an infinite amount of time.

Input

The first line of the input contains one integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains n ($1 \le n \le 2 \cdot 10^5$) — the number of stagecoaches. It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

The second line contains n integers $x_1, x_2, \ldots, x_n$ ($0 \le x_i \le 10^6$) — the positions of the stagecoaches. It is guaranteed that the positions are in strictly increasing order, so $x_i < x_{i+1}$ for $1 \le i \le n-1$.

The third line contains n integers $v_1, v_2, \ldots, v_n$ ($0 \le v_i \le 10^6$) — the speeds of the stagecoaches.

Output

For each test case, output a single integer, the number of stagecoach clumps that eventually form.

Example

standard input	standard output
2 2 10 12 3 2 3 10 12 14 2 4 3	1 2

Note

In the first test case, stagecoach 1 will catch up to stagecoach 2 eventually, leaving us with 1 clump.

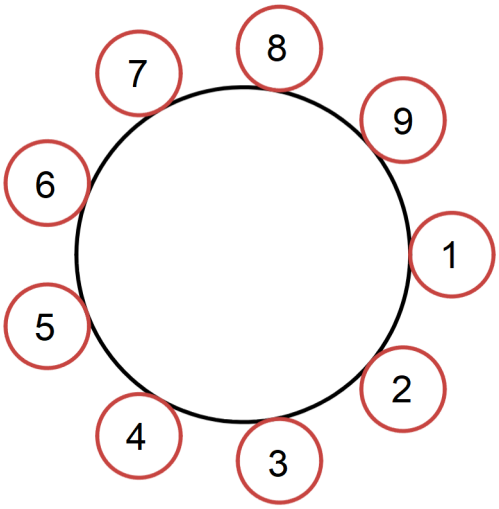
In the second test case, stagecoach 2 will catch up to stagecoach 3 eventually, but stagecoach 1 will never catch up with stagecoach 2. Therefore, eventually we will have clumps $\{1\}$ and $\{2, 3\}$, giving us 2 clumps.

Problem G. Dire Debt Dilemma

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

The saloon doors swing open, and a shady group steps in, wearing wide-brimmed hats and glaring eyes. Rowdy voices in the saloon quickly lower to a whisper. They make their way closer to your table, and your hands tighten and your eyes narrow. But the old sheriff next to you draws attention, clearing his throat and tapping his cane. Upon seeing his sheriff's gold star, the strangers walk away and sit down at a nearby table.

The n strangers are sitting around a circular table. Let them be labeled person $1, 2, \ldots, n$ going clockwise. Each person has either lent or borrowed money from their immediate left or right neighbors. Let p_i represent the money relationship that person i owes to the person to their left, person $(i \mod n) + 1$. If p_i is positive, it means person i owes p_i dollars to their left neighbor. If p_i is negative, it means their left neighbor owes person i p_i dollars. Each person has enough money to pay off all their debts. What is the minimum amount of money that can be transferred such that all debts are paid off? A person can only pay people sitting adjacent to them (left or right neighbors).



As an example, here is what a table with 9 people would look like. For example, person 9 could only transfer money to person 8 and person 1.

Input

The first line contains a single integer n $(3 \le n \le 10^5)$ — the number of people at the table.

The next line contains n integers $p_1, p_2, \ldots, p_n$ $(-10^4 \le p_i \le 10^4)$ — where p_i is the money relationship that person i has with their left neighbor.

Output

Output a single integer, the minimum number of dollars that can be transferred such that all debts are paid off.

Examples

standard input	standard output
4 10 9 8 7	4
3 1 1 -1	2
3 1 0 -1	2
5 -1 -7 -2 -4 -6	10

Note

In the first test case, person 1 owes person 2 \$10, person 2 owes person 3 \$9, person 3 owes person 4 \$8, and person 4 owes person 1 \$7.

The debts would all be settled if person 1 gave person 2 \$2, person 2 gave person 3 \$1, and person 1 gave person 4 \$1. The total amount of money transferred would be \$4.

In the second test case, person 1 owes person 2 \$1, person 2 owes person 3 \$1, and person 1 owes person 3 \$1.

The debts would all be settled if person 1 gave person 3 \$2.

Problem H. Tantilizing Tile Transformations

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

After enduring a mix of stagecoach and travel by foot, you arrive at the location of the gold mine on your map. But all you see in front of you is the face of a canyon. Dragging your hand across the canyon wall, you feel an array of small tiles embedded in the sandstone. Pressing the tiles causes the red, yellow, and orange stripes in the canyon wall to change. Crouching down, you also find a small inscription describing a strange pattern.

The tiles are arranged in an n by n grid of numbers $a_{i,j}$, where each number is in $\{0, 1, 2\}$. The inscription describes a pattern on an n by n grid that uses three colors: sandstone orange (0), desert yellow (1), or sunset red (2). Initially, the entire grid is the color of stone (0). You discover that by pressing the tiles, you can transform an entire row or column strip to the color of desert yellow (1) or sunset red (2). Find if it is possible to replicate the inscription's pattern on the grid. If so, provide a valid painting sequence.

Input

The first line of the input contains one integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \le n \le 1000$) — the width and height of the grid.

The i-th of the next n lines contains n integers $a_{i,1}, a_{i,2}, \ldots, a_{i,n}$ ($a_{i,j} \in \{0, 1, 2\}$) — the pattern on the grid.

It is guaranteed that the sum of n over all test cases is at most 10^5. It is also guaranteed that the sum of n^2 over all test cases is at most 10^6.

Output

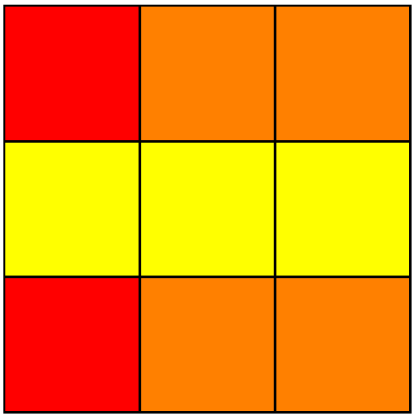
For each test case, output -1 if it is impossible to replicate the grid.

Otherwise, first output a single integer k ($1 \le k \le 8n$) — the number of presses needed to replicate the inscription. Then output k lines, where the i-th line contains three integers d_i, p_i, and c_i, where $d_i = 0$ if you transform the p_i-th row into the color c_i, and $d_i = 1$ if you transform the p_i-th column into the color c_i.

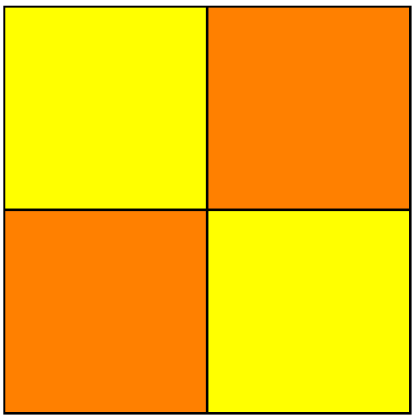
Example

standard input	standard output
3 3 2 0 0 1 1 1 2 0 0 2 1 0 0 1 1 0	2 1 1 2 0 2 1 -1 0

Note



In the first test case, shown above, the inscription can be replicated by first transforming column 1 to be red (color 2), then next transforming row to be 2 yellow (color 1).



In the second test case, shown above, it is impossible to replicate the inscription by pressing the tiles.

In the third test case, no actions are needed to replicate the inscription.

Problem I. Golden Cave Dash

Input file:	`standard input`
Output file:	`standard output`
Time limit:	3 seconds
Memory limit:	256 megabytes

Once the inscription is matched, the canyon walls open up to reveal a dark tunnel. It doesn't take long before the tunnel leads you to a gigantic cave glittered with gold!

Only one problem: walking down the tunnel, you notice footprints heading in, but none heading out. And worse yet, the footprints come in groups of three! You figure you can grab as much gold as you can before leaving the cave for good.

The cave can be modeled as a grid with n rows and m columns, with each cell containing some gold value $x_{i,j}$. Find the greatest common divisor of all the sums of cells on paths from the top left to bottom right that only go right and down.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains two integers n and m ($1 \le n \cdot m \le 2 \cdot 10^5$) — the number of rows and columns in the grid respectively.

The i-th of the next n lines contain m integers $x_{i,1}, x_{i,2}, \ldots, x_{i,m}$ ($1 \le x_{i,j} \le 10^9$) — the gold values in the grid.

Output

For each test case, output a single integer, the greatest common denominator of the sums over all paths.

Example

standard input	standard output
2 2 3 1 2 3 4 5 6 1 5 1 2 3 4 5	2 15

Note

In the first test case, there are 3 paths from the top left to the bottom right, and they have sums of $1+2+3+6 = 12$, $1+2+5+6 = 14$, and $1+4+5+6 = 16$. Therefore, the greatest common denominator over all paths is $\gcd(\{12, 14, 16\}) = 2$.

In the second test case, there is only one path from the top left to the bottom right, with sum $1 + 2 + 3 + 4 + 5 = 15$. Therefore, the greatest common denominator is $\gcd(\{15\}) = 15$.

Problem J. Rigging Revolver Rounds

Input file:	`standard input`
Output file:	`standard output`
Time limit:	3 seconds
Memory limit:	256 megabytes

This revolver has n chambers numbered 1 through n, each initially holding a numbered round of the same number. This revolver is a bit strange though, as instead of going around in a circle, there is instead a permutation p such that after firing round i, the next chamber aligned with the barrel is p_i.

Three-Legged Trigger likes to keep firing the revolver until he knows that whichever chamber he started with, the same chamber will be aligned with the barrel after.

You know this, but you don't have much time. With the time you have, you can do a single operation, where you choose two different chambers i and j and swap p_i and p_j in p. Find the single swap operation such that the number of firings that Three-Legged Trigger will do modulo $10^9 + 7$ is maximized.

Note that we are not finding swap that maximizes the number of firings, and taking that modulo $10^9 + 7$. We are finding the swap that has the largest residue modulo $10^9 + 7$.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains n ($3 \le n \le 2 \cdot 10^5$) — the number of chambers in the revolver. It is guaranteed that the sum of n over all test cases is at most $2 \cdot 10^5$.

The next line of each test case contains n integers $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$) — the permutation of chambers. It is guaranteed that all p_i are distinct.

Output

For each test case output three integers a, i, and j, where a is the number of firings modulo $10^9 + 7$ which is maximized, and i and j, $i \ne j$ are the chambers where the swap occurred.

If there are multiple solutions, print any of them.

Example

standard input	standard output
3 3 2 1 3 5 3 1 2 4 5 5 2 3 4 5 1	3 1 3 6 5 4 6 1 4

Note

In the first test case, swapping the values at $i = 1$ and $j = 3$ gives the permutation $p_1 = 3$, $p_2 = 1$, and $p_3 = 2$. It can be seen that after firing the revolver 3 times, it is guaranteed that whichever barrel Three-Legged Trigger started at will be aligned with the barrel.

In the second test case, swapping the values at $i = 5$ and $j = 4$ gives the permutation $p_1 = 3$, $p_2 = 1$, $p_3 = 2$, $p_4 = 5$, and $p_5 = 4$. It can be seen that after firing the revolver 6 times, it is guaranteed that whichever barrel Three-Legged Trigger started at will be aligned with the barrel.

Problem K. Scary Sagging Stalactites

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

You are exiting the cave when you hear the echoes of a triplet of footsteps grow louder and louder. You freeze. The Three-Legged Trigger has heard your footsteps, and he is coming! Above you are massive stalactites about to fall, which gives you an idea. If you cause the right stalactites to fall, you can block Three-Legged Trigger's path to you.

The cave can be modeled by a $n \times m$ rectangle of cells. Where the bottom left cell is $(0, 0)$ and the top right cell is $(n-1, m-1)$.

Three-Legged Trigger starts at the bottom left cell and wants to walk to the top right cell where you are standing in the cave. He is allowed to walk in any of the four cardinal directions (up, down, left, right). Furthermore, he cannot leave the rectangle at any time.

There are k stalactites, and when they fall, they form rectangular obstacles each defined by the bottom-left cell (blx_i, bly_i) and the top-right cell (trx_i, try_i). These obstacles **do not overlap**, but they can share edges. Find the minimum number of rectangular stalactites that need to fall to ensure Three-Legged Trigger cannot reach you no matter what path he takes? If it is not possible to trap the Three-Legged Trigger, output -1.

Note: It is guaranteed that none of the rectangles cover $(0, 0)$ or $(n-1, m-1)$.

Input

The first line contains three integers n, m, and k ($2 \le n, m \le 10^9$, $1 \le k \le 10^5$) — the width and height of the cave followed by the number of stalactites.

The next k lines describe the rectangles the stalactites create where the ith line contains four integers $blx_i, bly_i, trx_i, try_i$ ($0 \le blx_i, trx_i < n$, $0 \le bly_i, try_i < m$) which represent the bottom left and top right cells the rectangle covers. Furthermore, it is guaranteed that $blx_i \le trx_i$ and $bly_i \le try_i$.

Output

Output -1 if it is impossible to block the Three-Legged Trigger.

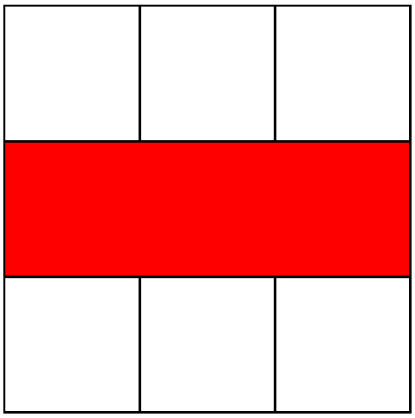
Otherwise, on a single line, output a single integer l ($1 \le l \le k$) — the minimum number of stalactites that needs to fall followed by l indices a_i ($1 \le i \le l, 1 \le a_i \le k$) of the chosen stalactites.

If there are multiple solutions, print any of them.

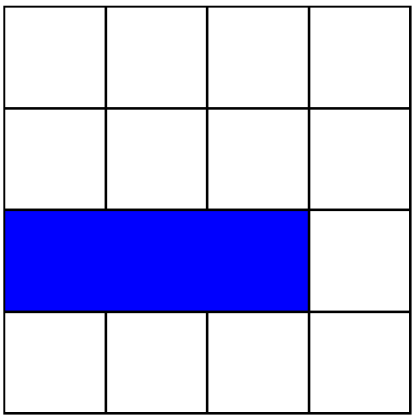
Examples

standard input	standard output
3 3 1 0 1 2 1	1 1
4 4 1 0 1 2 1	-1
4 4 2 0 1 2 1 3 1 3 1	2 1 2
4 4 4 0 3 0 3 1 2 1 2 2 1 2 1 3 0 3 0	4 1 2 3 4
4 4 5 0 3 0 3 1 2 1 2 2 1 2 1 3 0 3 0 2 2 3 2	3 1 2 5

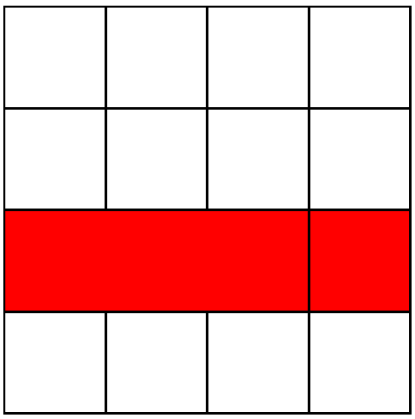
Note



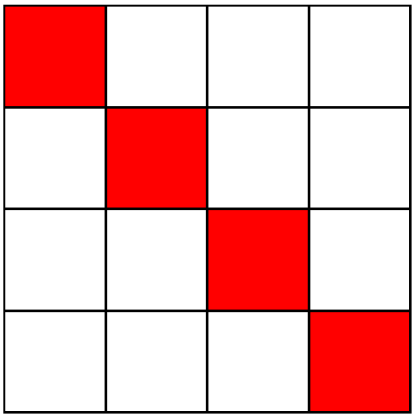
In the first test case, as shown above, we can block Three-Legged Trigger's path to you by making the stalactite fall.



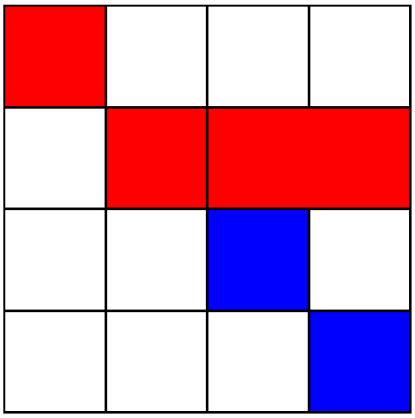
In the second test case, as shown above, we are unable to block Three-Legged Trigger's path to you.



In the third test case, as shown above, we can block Three-Legged Trigger's path to you by making both of the stalactites fall.



In the fourth test case, as shown above, we can block Three-Legged Trigger's path to you by making all of the stalactites fall.



In the fifth test case, as shown above, we can block Three-Legged Trigger's path to you in multiple ways. We can block it with the minimum number of stalactites possible by choosing rectangles 1, 2, and 5 shown in red above.

Problem L. Bye Bye! We out! - MAKING HAM KEK

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

Uhh... it looks like you may have knocked down too many stalactites because the entire cave is collapsing! You turn around and find yourself face-to-face with the barrel of a revolver.

The same group of strangers you saw at the saloon are standing in the background. And in front: the old deputy sheriff. His cane now has a boot attached at the bottom, and with his revolver aimed straight at your head, you are another victim of the infamous Three-Legged Trigger.

You slowly back up, glancing at the unstable stalactites above. The sheriff gives you one last respectful tip of his hat, and pulls the trigger. The world goes black.

You wake up under a pile of rubble. The Three-Legged Trigger and his gang are nowhere to be seen. It's time to drag yourself out of this cave for good.

The cave can be modeled as a weighted graph G. The paths in the cave that are not blocked can be modeled as a spanning tree T of G. Find all vertices v in the weighted graph G for which the shortest path weights from v to all other vertices, calculated using only the edges of the spanning tree T of G, are identical to the shortest path weights calculated using the entire graph G.

Input

The first line contains two integers n and m ($3 \le n \le 10^5$, $n - 1 \le m \le 2 \cdot 10^5$) — the number of vertices and the number of edges.

The next $n - 1$ lines contain three integers u_i, v_i, w_i ($1 \le u_i, v_i \le n$, $u_i \ne v_i$, $1 \le w_i \le 10^9$) — the edges in the spanning tree T where (u_i, v_i) is an edge with weight w_i.

The next $m - (n - 1)$ lines contain three integers u_i, v_i, w_i ($1 \le u_i, v_i \le n$, $u_i \ne v_i$, $1 \le w_i \le 10^9$) — the edges in G which are not in T, where (u_i, v_i) is an edge with weight w_i.

Output

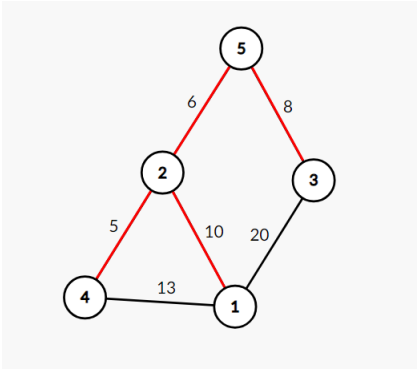
First output k, the number of vertices that satisfy the condition.

On the next line, output k space-separated integers, the vertices that satisfy the condition, in sorted order.

Example

standard input	standard output
5 6 1 2 10 2 4 5 5 2 6 5 3 8 1 4 13 1 3 20	2 2 5

Note



The first test case is shown above. The figure has edges in T in red and the edges not in T in black, combining to form G.

It can be seen that the shortest path from 2 to every other vertex using edges in T has the same weight as the shortest path from 2 to every other vertex using edges in G.

The shortest path from 1 to 4 using edges only in T has weight 15, while the shortest path using edges in G has weight 13, so 1 is not included in the answer.